

# Python Strings: A Reference

A string in Python is a sequence of characters enclosed in single ( `' '` ) or double ( `" "` ) quotes. Use this handout as a quick guide for core operations and methods.

## 1. Basic Operators & Functions

| Operator           | Description                                                    | Example & Output                                              |
|--------------------|----------------------------------------------------------------|---------------------------------------------------------------|
| <code>+</code>     | <b>Concatenation:</b> Joins two strings together.              | <pre>&gt;&gt;&gt; "Hello" + " " + "World" 'Hello World'</pre> |
| <code>*</code>     | <b>Duplication:</b> Repeats a string multiple times.           | <pre>&gt;&gt;&gt; "Python" * 3 'PythonPythonPython'</pre>     |
| <code>len()</code> | Returns the number of characters in the string.                | <pre>&gt;&gt;&gt; len("Apple") 5</pre>                        |
| <code>in</code>    | <b>Membership:</b> Checks if a character/substring is present. | <pre>&gt;&gt;&gt; "y" in "Python" True</pre>                  |

## 2. String Transformation Methods

| Method                  | Description                                | Example & Output                                                                |
|-------------------------|--------------------------------------------|---------------------------------------------------------------------------------|
| <code>.upper()</code>   | Converts all characters to uppercase.      | <pre>&gt;&gt;&gt; "hello".upper() 'HELLO'</pre>                                 |
| <code>.lower()</code>   | Converts all characters to lowercase.      | <pre>&gt;&gt;&gt; "HELLO".lower() 'hello'</pre>                                 |
| <code>.title()</code>   | Capitalizes the first letter of each word. | <pre>&gt;&gt;&gt; "python course".title() 'Python Course'</pre>                 |
| <code>.strip()</code>   | Removes whitespace from both ends.         | <pre>&gt;&gt;&gt; "  trim me  ".strip() 'trim me'</pre>                         |
| <code>.replace()</code> | Replaces a substring with another.         | <pre>&gt;&gt;&gt; "I like Java".replace("Java", "Python") 'I like Python'</pre> |

### 3. Search & Validation Methods

| Method                     | Description                                              | Example & Output                                           |
|----------------------------|----------------------------------------------------------|------------------------------------------------------------|
| <code>.find()</code>       | Returns the index of the first occurrence (or -1).       | <pre>&gt;&gt;&gt; "Hello".find("e") 1</pre>                |
| <code>.count()</code>      | Returns how many times a substring appears.              | <pre>&gt;&gt;&gt; "banana".count("a") 3</pre>              |
| <code>.startswith()</code> | Checks if the string starts with a specific value.       | <pre>&gt;&gt;&gt; "Python".startswith("Py") True</pre>     |
| <code>.isdigit()</code>    | Returns True if all characters are numbers.              | <pre>&gt;&gt;&gt; "123".isdigit() True</pre>               |
| <code>.split()</code>      | Splits a string into a <b>List</b> based on a separator. | <pre>&gt;&gt;&gt; "A,B,C".split(",") ['A', 'B', 'C']</pre> |